

# **STUDENT PROJECT TEMPLATE (SRS)**

## **Opticart**

### **Software Requirements Specification**

**Instructor:** Lara Nichols-Brown

**Term:** Spring 2026

**Section:** [Section Number]

#### **The Engineering Team:**

- **Student A:** Lucas Hom — Product Manager
- **Student B:** Samson Chang — Scrum Master
- **Student C:** Isaac Pressman — Lead Developer
- **Student D:** Amanda — Testing

#### **Project Assets:**

- **GitHub:** <https://github.com/samsonchang2028/csc-307-BigBallers>
- **Deployment:** [Link to Live Site]

#### **Document History:**

| Last date changed: | Who:  | What was changed:  |
|--------------------|-------|--------------------|
| 5/4/2026           | Lucas | User Stories added |
|                    |       |                    |
|                    |       |                    |
|                    |       |                    |

## **Opticart**

### **Software Requirements Specification (SRS)**

**Course:** CSC 307

**Date:** [Submission Date]

## **Product Vision:**

For grocery store customers who want cheaper prices for their groceries, Opticart is a general software that minimizes the cost of their grocery list by telling users which grocery store in their area is cheapest for their list. Unlike Google Shopping, our product allows users to compare prices of their entire grocery list at once.

## **User Stories**

### **Lucas:**

As a user, I want to enter my grocery list into the site so that I can find the grocery store that will minimize the price of my grocery list.

As a user, I want to have my grocery list saved on the site so that I don't have to enter it every time I want to check the minimum possible price of my groceries.

As a user, I want the ability to edit my saved grocery list so that when the items I require change, Opticart reflects this.

### **Samson:**

As a user I would like the prices of goods to be accurate so that when I buy the goods in person, I know I've optimized my purchase

As a user, I want to be instantly directed to the grocery list input so that I can take action on optimizing my grocery prices

As a user, I would like to log in so that I can have information saved for me

### **Isaac:**

As a user I want to see groceries near my location so I can optimize my trips

As a user I want to search certain groceries so I can get the best price

As a user I want to navigate through a map to see which grocery stores near me offer certain items

### **Amanda:**

As a user, I want to be able to quickly understand where I should input my grocery list and avoid spending too much time setting things up.

As a user, I'd like the ability to have a preference on which stores I'd like a comparison from.

As a user, I want to be able to easily add more items to my list without removing any of my previous inputs.

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### • Introduction

• **Problem Statement:** What specific problem are you solving?

• **Target Audience:** Who are the stakeholders?

• **scope:** what is the scope of this project?

• **user stories**

*User stories follow the format: "As a [Type of User], I want to [Action] so that [Value/ Benefit]."...*

• **US-01:** Authentication

• **US-02:** store Workout Log.

• **US-03:** search history

| ID    | Requirement                                                  | Priority |
|-------|--------------------------------------------------------------|----------|
| US-01 | As a user, I want to authenticate via username and password. | 2        |
| US-02 | As a user, I want to store workout information               | 1        |
| US-03 | As a user, I want to search workout history                  | 3        |
| US-04 | As a user, I want to add new workout types.                  | 4        |

### • Functional requirements

*The system shall...*

| ID    | Requirement                                                   | Priority |
|-------|---------------------------------------------------------------|----------|
| FR-01 | The system will authenticate users via username and password. | 2        |
| FR-02 | Allow users to store workout information                      | 1        |
| US-03 | Allow users to search their workout history                   | 3        |

|       |                                       |   |
|-------|---------------------------------------|---|
| US-04 | Allow users to add new workout types. | 4 |
|-------|---------------------------------------|---|

username and password.

|       |                                             |   |
|-------|---------------------------------------------|---|
| FR-02 | Allow users to store workout information    | 1 |
| US-03 | Allow users to search their workout history | 3 |
| US-04 | Allow users to add new workout types.       | 4 |

#### **4. Non-Functional Requirements**

- **Integrity:** handling errors and bad data
- **Security:** Implement bcrypt for password hashing and environment variables for DB credentials.
- **usability:**

#### **5. System Architecture**

- **REST Endpoints:** (List METHOD | URL | DESCRIPTION)
- **List your planned endpoints based on your REST slide:**
  - **GET /api/resources — Fetches collection.**
  - **POST /api/resources — Creates a new resource.**
- **UML:** (Attach a diagram)

- **User Interface (UI)**
  - **5.1 Wireframes / Mockups**
- **Data Requirements**

- **Traceability Matrix**

| ID    | Requirement                                                   | Line of code |
|-------|---------------------------------------------------------------|--------------|
| US-01 | The system will authenticate users via username and password. | 152          |
| US-02 | Allow users to store workout information                      | 256          |
| US-03 | Allow users to search their workout history                   | 46           |
| US-04 | Allow users to add new workout types.                         | 45           |

- **AI Usage & Disclosure (Mandatory)**
- **Model(s) Used:** [e.g., Claude 3.5, GPT-4o]
- **prompts used during coding:**